systems

MDPI

Article

Advancing Dynamic Emergency Route Optimization with a Composite Network Deep Reinforcement Learning Model

Jin Zhang [1], Hao Xu [1], Ding Liu [1] and Qi Yu [2,*]

[1] College of Transport & Communications, Shanghai Maritime University, Shanghai 201306, China; 185473@shmtu.edu.cn (J.Z.); 202330610042@stu.shmtu.edu.cn (H.X.); dliu@shmtu.edu.cn (D.L.)
[2] College of Transportation Engineering, Tongji University, Shanghai 200092, China
* Correspondence: 2233440@tongji.edu.cn

Abstract: Emergency logistics is essential for rapid and efficient disaster response, ensuring the timely availability and deployment of resources to affected areas. In the process of rescue work, the dynamic changes in rescue point information greatly increase the difficulty of rescue. This paper establishes a combined neural network model considering soft time-window penalty and applies deep reinforcement learning (DRL) to address the dynamic routing problem in emergency logistics. This method utilizes the actor–critic framework, combined with attention mechanisms, pointer networks, and long short-term memory neural networks, to determine effective disaster relief path, and it compares the obtained scheduling scheme with the results obtained from the DRL algorithm based on the single-network model and ant colony optimization (ACO) algorithm. Simulation experiments show that the proposed method reduces the solution accuracy by nearly 10% compared to the ACO algorithm, but it saves nearly 80% in solution time. Additionally, it slightly increases solution times but improves accuracy by nearly 20% over traditional DRL approaches, demonstrating a promising balance between performance efficiency and computational resource utilization in emergency logistics.

Keywords: emergency logistics; dynamic route optimization; deep reinforcement learning; composite network; vehicle routing problem

check for updates

Academic Editor: Mahyar Amirgholy

Received: 3 January 2025
Revised: 8 February 2025
Accepted: 14 February 2025
Published: 17 February 2025

Citation: Zhang, J.; Xu, H.; Liu, D.; Yu, Q. Advancing Dynamic Emergency Route Optimization with a Composite Network Deep Reinforcement Learning Model. *Systems* **2025**, *13*, 127. https://doi.org/10.3390/systems13020127

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

With the continuous advancement of technology, the prevention techniques for various natural disasters and the ability of human society to resist disasters have become more mature. However, after an emergency has happened, the impact on people's daily activities cannot be ignored, for example, explosions in factories and traffic accidents due to some accidents. Therefore, the government has always attached great importance to the rescue work, so that emergency logistics have been developed. Emergency logistics refers to the unconventional logistics form that ensures the availability of emergency supply in the abnormal state of emergencies. This concept has been proposed since 1984 [1]. The development of emergency logistics is a guarantee for protecting national interests, maintaining social stability, and enhancing national security [2].

At present, with the advancement of technology, the application of intelligent logistics information systems, cloud computing, the Internet of Things, big data analysis, artificial intelligence, and other intelligent technologies has gradually penetrated into the logistics industry, including emergency logistics [3]. In the process of emergency relief, the intelligent logistics information system can monitor the transportation of emergency supplies, monitor the location of emergency vehicles and other data in real time, and quickly determine the

demand and the best transportation path [4]. The Internet of Things technology can monitor the temperature and humidity of the supplies and other information to ensure the safety and reliability of the transportation process. Through big data analysis technology, historical and live data can be analyzed to forecast the scope of the impact of disasters, optimize the emergency supply reserves and scheduling solutions, and help the emergency sector better understand the needs of the disaster area and improve response efficiency. Artificial intelligence technology can be applied to route planning, material dispatching, and demand forecasting. Machine learning algorithms enhance the accuracy of supply–demand predictions following a disaster. The comprehensive use of these intelligent technologies can improve the operational efficiency of emergency logistics, optimize the material dispatching program, help rescue teams better respond to emergencies, and safeguard the life safety and basic needs of disaster-affected people [5,6].

After an emergency has happened, emergency supplies need to be deployed to the disaster area in a short period of time. However, rescue work faces many challenges under various complex conditions, resulting in low efficiency and high losses [7]. The widely concerning and discussed issue among scholars in the field of emergency logistics is to take an effective rescue work to avoid causing a wider loss. The current academic research in the field of emergency logistics mainly includes emergency supply–demand forecasting, emergency stockpile management, emergency logistics center site selection, emergency logistics scheduling, and so on. Among those problems, emergency logistics scheduling plays an essential role [8].

Emergency vehicle routing is an effective method used in the process of emergency logistics scheduling, and it is also an important factor that affects the rescue efficiency. Vehicle Routing Problems (VRPs) are widely applicable in areas like logistics distribution, public transportation, and garbage collection. Optimizing vehicle paths can improve logistics efficiency, reduce logistics costs, and so on [9]. For the emergency logistics problem, the traditional VRP cannot be adapted to the complex reality; for instance, the rescue point information can change dynamically in emergency situations. Therefore, the dynamic VRP (DVRP) is the focus of consideration in the emergency logistics problem [10]. DVRP refers to the problem of how to plan and optimize the path of logistics vehicles under dynamic constraints such as the uncertainty of supply–demand, the change in the location of the demand point, and the cargo load situation [11]. In emergency logistics, due to the fact that rescue information may change, emergency vehicles need to transmit the real-time location of the vehicles and rescue changes to the emergency logistics dispatch center in a timely manner through intelligent technologies, such as the Internet of Vehicles, and the dispatch center then uses intelligent technologies to transmit the scheduling plan suitable for the current situation to the rescue team. The traditional static vehicle path problem cannot adapt to the complex reality, so the DVRP is more realistic in emergency logistics research.

The computational complexity of the VRP is well-documented in the literature, classified as NP-Hard based on its transformation from the Traveling Salesman Problem (TSP). This classification is derived from pivotal studies: In 1972, Karp demonstrated the NP-Hardness of TSP, a foundational result in computational complexity theory [12]. Building on this, Lenstra and Kan, in 1981, effectively mapped VRP to a known TSP configuration, thereby confirming the NP-Hard status of VRP [13]. NP-hard problems usually contain numerous combinations and permutations of high computational complexity, making the search for an optimal solution difficult or even infeasible, and in practice, exact algorithms have unacceptable complexity for this type of problem. Therefore, NP-hard problems are usually solved using heuristic algorithms or other methods to obtain a suboptimal or approximate solution in an acceptable time. In the VRP addressed in this study, under the consideration of dynamic changes in the number of rescue points, dynamic information

needs to be transmitted to the emergency logistics dispatch center in time to make timely decisions. Traditional algorithms may not be able to obtain the ideal strategy within a certain time in the face of the changing number of rescue points. DRL, on the other hand, attains the strategy by interacting with the environment, which makes it extremely adaptable in the face of dynamic environment [14]. When the number changes, DRL continuously adapts its strategy to the new rescue scale through the interaction between the intelligence and the environment. Therefore, this paper focuses on VRP with time window (VRPTW), establishing a Bayesian-weighted combinatorial neural network model; and solving the constructed model using DRL algorithms to obtain a better emergency logistics scheduling solution under the change in rescue scale to optimize the cost, improve efficiency, and promote the coordinated development of intelligent emergency logistics.

2. Literature Review

The concept of VRP was initially introduced by Dantzig and Ramser [15]. VRP is a combinatorial optimization problem aiming to obtain the shortest path to reach the destination that satisfies all customer demands and minimizes the total distance or the total cost under some constraints. Balinski et al. [16] argued that research for VRP can be inserted from the perspective of modeling and algorithms. For the modeling, most of the research in recent years has added suitable constraints to VRP to meet the increasingly rich scenarios. For example, in response to the national low-carbon environmental protection policy, Zhang et al. [17] considered factors such as government regulatory policies and firm regulatory investments to determine vehicle paths. They proposed an improved genetic algorithm (GA) and proved its effectiveness. Wang and Wen [18] proposed a VRPTW model by aiming to minimize total cost and carbon emission cost and maximize customer satisfaction. They also introduced an adaptive GA to solve the model. Amiri et al. [19] explored a Green Vehicle Routing Problem (GVRP) for a mixed fleet of conventional and electric trucks, and they developed a VRP model with the objective of minimizing the cost and greenhouse gas emissions. In real-world scenarios such as logistics distribution and courier services, customers usually can only receive services within a specific time window. Therefore, the VRP model considering time-window constraints has gradually become the focus of research. Jiang et al. [20] introduced a multi-mode hybrid electric VRPTW (MM-HEVRPTW), which optimizes both the route selection and the operational mode for each road segment, taking into account factors like battery capacity and customer time windows. In order to meet the demands of distribution under challenging practical conditions, Jie et al. [21] developed a time-dependent VRP with a soft time-window model, using the minimization of total distribution cost as the objective function, and also proposed a HA to solve the model. In the practical route-planning process, factors such as customer demand, service time, and traffic conditions may vary over time. Therefore, it is essential for VRP to account for these dynamic changes in order to optimize strategies effectively. In order to raise construction waste transportation efficiency and reduce carbon emission, Wang et al. [22] addressed the dynamic time-dependent GVRP model that incorporates factors such as urban traffic congestion, time-varying speeds, and those influencing carbon emissions. They proposed a stochastic sampling method to tackle uncertain customer requests. Considering the uncertainty of information in real scenarios, it is necessary to incorporate dynamic constraints into the VRPTW. Whether deliveries can be completed within the designated time window is a key indicator of urban logistics efficiency. Moreover, the rapid growth of the urban logistics recycling industry has created a surge in dynamic demands. Therefore, Wang et al. [23] considered the time window and dynamic customer demand constraints to establish a bi-objective model that minimizes the operating cost and the number of vehicles. The background of the increasing demand for emergency response

has led to a growing number of research studies on emergency VRP (EVRP). Zhang et al. [4] explained the importance of emergency logistics after emergencies, analyzed the features of emergency logistics, established a multi-objective location optimization model based on a node-weighted bottleneck Steiner tree, and proposed an intelligent algorithm using cellular stochastic diffusion search to solve the problem. The dynamism of emergency logistics is highlighted in their article. Dynamism is a key characteristic of EVRP, making traditional static VRP unsuitable for this scenario. Najafi et al. [24] constructed a dynamic vehicle-scheduling model considering the transportation scheduling of supplies and casualties during the earthquake, with the goal of minimizing the scheduling time. They conducted experiments to analyze how network topology impacts the speed of earthquake response, aiming to enhance the overall quality of response efforts. For the purpose of addressing the dynamism of transportation in disaster response, Peng et al. [8] reformulated the truck–drone collaborative routing problem as a Markov game model based on an event-driven approach. This model effectively captured the dynamic changes in the states and node information of trucks and drones during the resource process.

Existing research studies show a shift in VRP models from single-objective to multi-objective problems that consider various complex constraints, particularly time windows and dynamic constraints. In the context of the EVRP studied in this paper, where information changes over time and rescue points are highly dependent on response times, time window and dynamic constraints become the core of this study.

Research on algorithms for solving VRP has been continuously optimized and improved as the problem progresses. Early scholars mainly focused on the exact algorithm. For example, Ceselli et al. [25] proposed the VRP during the optimization of a waste management system. They presented an exact optimization algorithm and proved experimentally that the algorithm is superior in solving the real problem within a reasonable computation time. Lozano et al. [26] designed an exact solution method based on the hidden enumeration approach to solve the shortest path problem with resource constraints, which can greatly reduce the search space during computation. Compared to exact algorithms, heuristic algorithms are more capable of obtaining a suboptimal or better solution in an acceptable timeframe by reducing the accuracy in exchange for the speed of solving the problem [27]. Li et al. [28] studied TDVRPTW with spatiotemporal distance (TDVRPTW-TSD), with the goal of minimizing total travel time and maximizing customer satisfaction while meeting vehicle capacity. They developed an efficient local search heuristic algorithm combining adaptive large neighborhood search (ALNS) and variable neighborhood descent (VND) to improve the exploration ability, and they compared it with several algorithms to verify the efficiency and effectiveness. Considering the variation in information, such as customer demand, Salamatbakhsh-Varjovi et al. [29] established a Periodic Vehicle Routing Problem (PVRP) with the objectives of minimizing transportation costs and maximizing the expected value of satisfying customer demand. They used the Improved Differential Evolution (IDE) algorithm to solve the model, demonstrating its accuracy and effective performance. The ACO algorithm, as a classical heuristic algorithm, is widely used to solve VRP due to its excellent adaptability and global search capability. In the existing literature on solving VRP using the ACO algorithm, most researchers have made modifications to the traditional ACO algorithm or combined it with other algorithms to enhance its performance. For example, Li et al. [30] developed an improved ACO algorithm (IACO). By optimizing the updating of pheromone, the search range of the algorithm becomes larger. Compared to the traditional ACO algorithm, IACO has a higher solution quality and shows good responsiveness and simplicity for VRP. Considering the complexity of the VRP involving simultaneous pickup–delivery and time window (VRPSPDTW), based on the ACO algorithm, Wu et al. [31] established an ACO algorithm with a destroy-and-repair

strategy (ACO-DR). The experimental results show that the algorithm has utility in real-world problems. This method incorporates a directional random transition rule to improve global search efficiency and adds two local operators—destroy and repair—to reduce the risk of convergence to local optima. Experimental findings validate that the algorithm has utility in real-world problems. Teng et al. [32] proposed an improved evolutionary algorithm based on Ant Colony System. Experiments show that the algorithm converges quickly, and its ability to solve dynamic problems is demonstrated in example validations. Zhang et al. [7] proposed a hybrid algorithm (HA) that combines artificial immunity and ACO to calculate the proposed grain EVRP with three objectives.

With the gradual maturation of the development of intelligent logistics, intelligent technologies such as DRL have also been applied to the field of emergency logistics and have been widely used in EVRP. DRL algorithms have higher search efficiency than heuristic algorithms and have the advantages of parallelism, generality, and globalization [33]. Chung [6] provided a thorough review of the applications of smart technologies (STs) in improving transportation efficiency. They discussed the technical difficulties encountered in developing ST applications. Tang et al. [34] proposed an end-to-end Transformer-based DRL method to efficiently determine energy-optimal routes for fleets of electric logistics vehicles to minimize operating costs. Xin et al. [35] proposed an innovative step-by-step scheme which explicitly removes visited points at every selection step. They applied this scheme to attention mechanism and pointer network and markedly improved the performance. Existing DRL approaches often overlook service-time constraints, which require vehicles to remain at current nodes until service times are met, limiting their real-world applicability. Therefore, Wang et al. [36] introduced heterogeneous VRP with service-time constraints (HVRP-STC) and formulated it as a Markov Decision Process (MDP) incorporating these constraints. They developed an innovative DRL model to solve this problem. Zhang et al. [37] developed a new DRL algorithm, the Multi-Agent Attention Model, to solve the VRPTW. Additionally, they developed a multi-agent DRL approach incorporating an unsupervised auxiliary network for the model training. The results demonstrated that the algorithm solves faster than traditional algorithms. With the continuous development of DRL algorithms, their application in DVRP is gradually becoming a mainstream approach. Secomandi [38] used neuro-dynamic programming (NDP) to solve the VRP with uncertain customer demands. The study compared two algorithms—optimistic approximate policy iteration and a rollout policy. The results showed that while the former improved the nearest-neighbor policy's performance by 2.3%, the rollout policy produced higher-quality solutions. Meanwhile, Zhou et al. [39] used a different approach. They used the neural network to learn the basic policy, utilized a reinforcement learning framework to enable the basic policy to explore new scenarios, and then applied the newly generated data to further train the basic policy as a means of solving the VRP with uncertain customer demands.

From the existing studies, it can be seen that the algorithms for solving VRPs have gradually evolved from exact algorithms to various improved heuristics algorithms. With the development and application of artificial intelligence, DRL, which can derive strategies through interaction with the environment so that it can effectively adapt to the environment, has been gradually applied to VRP, particularly in problems involving time windows and dynamic constraints. However, the existing literature using the improved ACO algorithm does not take into account the situation that the number of points dynamically changes during the distribution process, and the studies using the DRL algorithm to solve this problem lack the modeling of uncertainty, as well as the poor generalization ability of the model. Therefore, in the VRP considered in this study, the ACO algorithm may fail to identify the optimal strategy in a given time due to the dynamic variation in the number of rescue points, although it possesses excellent adaptability and global search capability.

In order to provide a scientific path strategy for the rescue team, this paper establishes a combinatorial neural network model considering the soft time-window penalty, applies the DRL algorithm to solve the dynamic EVRP, and compares its results with the ACO algorithm and DRL algorithm based on the traditional single-critic network.

3. Deep Learning-Based Combinatorial Neural Network Logistics and Transportation Scheduling Model

3.1. Modeling of Logistics Scheduling Problem Considering Time-Window Penalty, Transportation Cost, and Dynamic Demand Under Single-Network Prediction

3.1.1. Problem Description and Modeling

In a logistics dispatch system, the distribution center dispatches vehicles to satisfy the customer demand. This study focuses on emergency relief, where the "customer" is defined as the rescue point targeted in emergency logistics operations. During transportation, the number of rescue points can change dynamically. In response, the disaster relief team needs to rapidly adjust to changes in the delivery route to minimize costs. This study proposes a DRL method that combines an actor–critic model and a pointer network [5] to solve the vehicle-scheduling problem with dynamically changing information about relief points during disaster relief. The method architecturally utilizes the actor–critic framework and combines the pointer network to determine an effective disaster relief path sequence. The actor explores the stochastic path sequences through simulation and evaluates the total cost using a costing function based on MDP [40]. LSTM [41], the core of the critic network, is trained to accurately predict the value of each step of the relief team. By integrating the trained critic network into the actor decision process and applying the attention mechanism, the action steps evaluated as optimal are selected.

Model assumptions: Before modeling the research problem, the following assumptions are made in this paper to simplify the disaster relief problem:

- The coordinates of the location of the emergency supply center, as well as the relief point, are known;
- The loading capacity of the relief vehicles is large enough;
- The vehicle starts from one emergency center and eventually returns to the same emergency center;
- There is one and only one disaster response team available for each relief point;
- There are no transportation routes between emergency response centers;
- Traffic conditions on the roads were not affected;
- If a vehicle arrives before the relief time window, it will be rewarded; conversely, it will be penalized for exceeding the time limit.

This problem can be modeled as an MDP that can be solved using reinforcement learning.

3.1.2. Initialization Information

Due to the deep learning method of the neural network applied in this paper, the current information of the disaster relief team and the status of the rescue point are represented by a one-dimensional array Equation (1):

$$Z_t = \{C_t, T_t, N_1, N_2 \ldots N_n\} \tag{1}$$

where Z_t denotes the state information of the rescue point and the disaster relief team at moment t; C_t denotes the load that the disaster relief team is taking on at moment t; and T_t denotes the absolute time of the relief team at moment t, which is used to calculate the

delay penalty and early reward for the arrival of the relief team. In the initial state, the absolute time, T_0, is 0. $N_1, N_2 \ldots N_n$ denote the demand of the n th point at moment t.

The initial carrying capacity, C_0, can be obtained by the known total demand, which is calculated as Equation (2):

$$C_0 = \sum_{i=0}^{n} N_i \tag{2}$$

For each rescue point, it contains the demand, rescue demand time, and coordinate location, which is represented as $(X_i, Y_i, N_i, T_i, A_i)$ in a one-dimensional array. Here, X_i and Y_i denote the horizontal and vertical coordinates of the i th rescue point on the map, respectively; N_i denotes the demand for the i th rescue point; and T_i denotes the rescue cut-off time for the i th rescue point.

In this study, the uniform random distribution is used to obtain the initial coordinates of the rescue points during training, and then the normal distribution is used to attach the rescue demand time and demand to each rescue point. However, in the simulation verification process, the number of rescue points is dynamic, and it is necessary to increase the rescue generation time by using the absolute time, T_t, which can be expressed as $(X_i, Y_i, N_i, T_i, A_i)$, where A_i denotes the generation time of the rescue point.

3.1.3. State Transfer Function

In the dynamic emergency logistics scheduling problem, the state vector of the current situation is represented by a one-dimensional array of length n + 2, with n being the number of all known rescue points. When the latter n items of this array are all 0, there is no demand point that still needs to be rescued. It is judged to be the completion of this rescue, and the disaster relief team returns to the center.

When the rescue proceeds to moment t, the state transfer function is as shown in Equation (3):

$$Z_0 = \{C_0, T_0, 0, 0, 0 \ldots 0\} \tag{3}$$

At this point, there are n-t rescue points left, and the value of the next drive to each of the remaining points $\{R_1, R_1, \ldots R_{n-t}\}$ can be obtained from the critic network.

After the actor takes the next step, t is increased by one, and the system needs to update this state:

$$Z_{t+1} = \{C_{t+1}, T_{t+1}, 0, 0, \ldots, 0, N_{k+1}, \ldots N_n\} \tag{4}$$

$$C_{t+1} = C_t - N_{k+1} \tag{5}$$

$$T_{t+1} = T_t + \frac{d_{k+1}^k}{V} \tag{6}$$

$$N_{k+1} = 0 \tag{7}$$

The current arrival point is k + 1. Equation (4) represents the state of the relief team at the moment t + 1; Equation (5) represents the amount of supplies remaining in the relief team at the moment t + 1; Equation (6) represents the current absolute time increase in the time taken by the relief team to drive from the k th point to the k + 1 th point; and Equation (7) represents the amount of demand at the visited point that is satisfied. V is the traveling speed of the team, and d_{k+1}^k is the Euclidean distance from the k th point to the k + 1 th point:

$$d_k^{k+1} = \sqrt{(X_k - X_{k+1})^2 + (Y_k - Y_{k+1})^2} \tag{8}$$

In the simulation experiments, a rescue point update mechanism is added to emulate the dynamic change in rescue point size. Demand generation time for new rescue points is $A_i \in [T_t, T_{t+1}]$. The value of the next drive to each of the remaining points is $\{R_1, R_1, \ldots R_{n-t}, R_{n-t+1}\}$, and the formulated demand for the new rescue point, N_i', is assigned to the new state sequence, $Z_{t+1} = \{C_{t+1}, T_{t+1}, 0, 0, \ldots, 0, N_{k+1}, \ldots N_n, N_{n+1}\}$ in N_{n+1}, as shown in Equation (9):

$$N_i = N_i' \tag{9}$$

3.1.4. Reward Function

The DRL used in this chapter is data-driven. Without data, initial data are generated from this by setting up environmental reward rules and randomly simulating the initial information. The rescue sequence $\{0, 1, 2 \ldots k \ldots 30\}$ obtained by decoding is the input to the reward function. Since this paper is a VRPTW that comprehensively considers the time window and shortest path, the optimization of rescue cost represents a combination of total path length, capacity cost, and delay cost, so the reward function is as follows:

$$\sum_{i=0}^{n-1} d_i^{i+1} + \alpha \times \sum_{i=0}^{n-1} \left(C_0 - \sum_{j=0}^{i-1} N_j \right) \times d_i^{i+1} + \beta \times \sum_{i=0}^{n} \left(T_i - \frac{\sum_{j=0}^{i-1} d_j^{j+1}}{v} \right) \tag{10}$$

where $\sum_{i=0}^{n-1} d_i^{i+1}$ denotes the total distance of the path through n rescue points; $C_0 - \sum_{j=0}^{i-1} N_j$ denotes the residual amount of vehicle capacity when the relief team travels to the i th point; α is the transportation cost coefficient; and $\sum_{j=0}^{i-1} d_j^{j+1}$ denotes the distance traveled by the relief team to proceed to rescue point i. Dividing it by V gives the time taken by the rescue team to get to the rescue point i. Subtracting the rescue demand time at point i from the total time taken to reach point i gives the time-window delay penalty and early reward. β is the coefficient of the time delay penalty and early reward. In a traditional VRPTW, only one-sided overdue delays are generally used without early rewards. However, in this paper, the rewards from early rescue need to be considered in a post-disaster scenario.

3.1.5. Actor–Critic Modeling Based on a Single LSTM

As a deep learning tool, neural networks need to be given a large amount of input and output data. In this subsection, by analyzing the information such as the coordinates of rescue points and their waiting time for rescue, these data are made into a one-dimensional array, which is used as an input to the neural network. By using the actor–critic model in DRL, a seq2seq model framework based on pointer network has been designed to solve the disaster relief problem under dynamic demand scenarios. The structure of the actor–critic framework is shown in Figure 1:

The framework consists of three parts:

(1) Actor receives a one-dimensional array, Z_t, of the current state and computes the actual reward from the predicted path of the critic network according to the rule.
(2) The critic network judges the value of each step of the behavior based on its own output and calculates the probability of going to the next step using the attention mechanism.
(3) Decode the sequence based on the output of the attention network and compute the array of next states.

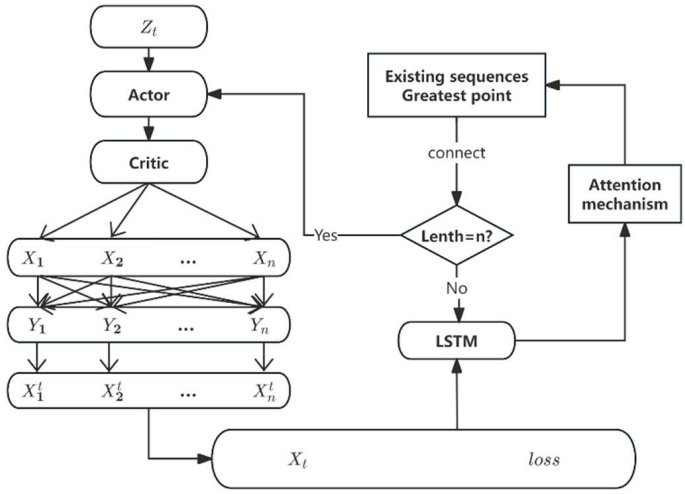

Figure 1. The actor–critic framework.

3.1.6. Attention Mechanism

This paper introduces an attention mechanism to handle the network input of large-scale rescue points and solve the value and access probability of each rescue point:

$$P_i = \frac{Reward_i}{\sum_{i=0}^{n} Reward_i} \tag{11}$$

where $Reward_i$ denotes the value toward the i th point, and the probability of going toward this point is calculated as the ratio of its value to the total value of the all currently feasible rescue points. This results in a sequence of probabilities of moving toward each point: $P = \{P_1, P_2, P_3 \ldots Pi \ldots P_n\}$. The authors output the hidden states of LSTM as a sequence of a, and then they perform softmax normalization with the similarity between a and each hidden state to obtain the attention weight, b. The probability of going to each rescue point can be obtained by multiplying b and the output of the hidden states with the following formula:

$$b = \theta \tanh(\eta[a]) \tag{12}$$

$$c = softmax(b) * \{P_1, P_2, P_3 \ldots P_i \ldots P_n\} \tag{13}$$

where θ and η are coefficients, and tanh and softmax are activation functions.

3.1.7. Critics Network

In this paper, LSTM is used to estimate the critic value. In this network, there are structures such as forgetting gates, input gates, and output gates. Forgetting gates combine the output of the previous cell with the input of the current cell (the cell can also be interpreted as an iteration) and then output to the next cell by the sigmoid function.

The training set contains the sequence of relief paths, the distribution of the relief points, the demand time, and the supply–demand for each relief point. Expanding all the training data one-dimensionally using the reshape function, a one-dimensional array can be obtained and inputted into the critic network to finally obtain the estimated value of each feasible rescue point.

3.1.8. Deep Reinforcement Learning-Based Algorithmic Process for Logistics Scheduling

The steps of the scheduling algorithm based on DRL are shown in Algorithm 1:

Algorithm 1. The steps of the logistics scheduling algorithm based on DRL.

```
Initializing the critic network
Initialize training set parameters
For each training set loop:
Generate N training sets
For each training set loop:
    Split training set into samples based on training batch
      For each sample loop:
          Initialize state function, Z0
          Loop:
Actor generates actions and computes the true reward
Critic estimates the reward and generates the sequence by decoding it
                End loop
            Calculate loss and train the network
End loop
End loop
Input test set
Input the requirement table according to the filtered update mechanism of the tests
Loop time, Tt:
Input the remaining demand sequences into the neural networks
Weighted average of results from each neural network
End the loop
```

3.2. Modeling of Logistics Scheduling Problems Considering Time Windows and Transportation Costs in the Case of Combinatorial Neural Networks

3.2.1. Feature Modeling of Sub-Networks in Combinatorial Networks

During the training of the critic networks in the actor–critic model, a one-dimensional array dataset of length 5 is generated using the same rules for the training of each unique network:

$$Rule_i = \{minCN_i, maxCN_i, minCT_i, maxCT_i, CG_i\} \tag{14}$$

where $Rule_i$ denotes the i th dataset generation rule.

3.2.2. Weighted Average of Sub-Networks in Combinatorial Networks

Based on the actor–critic framework under DRL, when the actor is at any non-terminal node, there will be several feasible nodes. A critic network based on different generation rules will predict the value of each node. From this, a value matrix is generated, which is represented by the following Equation (15):

$$TReward_{i,t} = \begin{bmatrix} Reward^1_{1,t} & Reward^1_{2,t} & \dots & Reward^1_{m,t} & \dots & Reward^1_{n-t,t} \\ Reward^2_{0,t} & \dots & \dots & Reward^2_{m,t} & \dots & Reward^2_{n-t,t} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & Reward^j_{m,t} & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ Reward^J_{1,t} & Reward^J_{2,t} & \dots & Reward^J_{m,t} & \dots & Reward^J_{n-t,t} \end{bmatrix} \tag{15}$$

where $TReward_{i,t}$ denotes the value matrix of an actor traveling from the current i th rescue point to the remaining feasible rescue points at moment t; and $Reward^j_{m,t}$ denotes the value of an actor traveling from the i th rescue point to the m th rescue point at moment t under

the prediction of the critic network based on the j th generation rule. The size of this matrix is $J \times (n - t)$, where J is the number of generation rules, and n is the number of rescue points. Since t moments have elapsed and t rescue points have been traveled, the number of remaining rescue points is n − t.

During weighted averaging, for each critic network under a different generation rule, it needs to be multiplied by the corresponding weight, e_j. All e_j values together form a one-dimensional array of length J:

$$E = \{e_0, e_1, e_2, \ldots, e_j, \ldots e_J\} \tag{16}$$

$$\sum_{J}^{j=1} e_j = 1 \tag{17}$$

For the purpose of facilitating the multiplication of this array with the value matrix, the array is expanded into the following matrix:

$$E = \begin{bmatrix} e_0 & 0 & \ldots & \ldots & 0 & 0 \\ 0 & e_1 & \ldots & \ldots & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & e_j & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots & e_{J-1} & 0 \\ 0 & 0 & 0 & 0 & 0 & e_J \end{bmatrix} \tag{18}$$

Then, the weighted reward matrix is represented as Equation (19):

$$AReward_{i,t} = E \times (TReward_{i,t})^T \tag{19}$$

where $(TReward_{i,t})^T$ denotes the transpose matrix of $TReward_{i,t}$.

The weighted value expectation of the remaining points is obtained by summing each vertical column of $AReward_{i,t}$. The feasible point with the largest value is selected to determine the next optimal action.

3.2.3. Solving for Combinatorial Network Weights

For a critic network that generates a training set based on a specific generative rule, a weight, e_j, is generated that is influenced by the feature parameters of the real situation. In the relationship between the characteristic parameters of the real situation, the characteristic parameters of the critic network and the weights, the two characteristic parameters are independent variables with the number of 10. The weights are a single dependent variable, so this weight can be obtained by the method of multiple linear regression.

Setting the characteristic parameters of the real situation and the critic network as y_1 to y_{10}, and the parameters of the multiple linear regression as M_1 to M_{10}, the relationship between the three can be expressed as Equation (20):

$$e_j = M_1 \times y_1 + M_2 \times y_2 + \ldots + M_i \times y_i + \ldots + M_{10} \times y_{10} \tag{20}$$

By fitting the multiple regression coefficients through a large number of cases, a method for calculating the weights of the predicted results of different critic networks in the combined network can be obtained.

The weights need to satisfy the condition $\sum_{j=1}^{J} e_j = 1$. The weights at this time are summed, and $\sum_{J}^{j=1} e_j$ is recorded. For each e_j, the following equation is performed:

$$e_j = \frac{e_j}{\sum_{j=1}^{J} e_j} \tag{21}$$

In summary, this section improves model performance by combining multiple critic networks adapted to the current scenario with a weighted average of predicted values.

4. Case Study

In this section, simulation experiments are conducted to evaluate the proposed model.

4.1. Data Description

This subsection simulates scenarios involving 20, 40, 60, and 100 standard and dynamically growing scale points. Initially, 200,000 instances are generated as the training set, each of which is a single-vehicle transportation problem with a specific rescue size and dispatch center. During training, 200 rescue points are selected from the training set at a time, and the dispatch center node is randomly added as the starting node. The location of the dispatch center node is distributed in the same way as the rescue point by setting its time window to [0, 100] without considering the demand of the dispatch center. The data for rescue point, i, of the validation and test sets are represented using $X_i = (x_i, y_i, T_i, N_i)$. The demand, N_i, for each rescue point is chosen probabilistically and randomly from a discrete number $\{0, 1, 2, 3, 4, 5, \cdots\cdots, 40\}$. The locations (x_i, y_i) of rescue points and dispatch centers are randomly generated in a city modeled by a planar space with dimensional area $[0, 100] \times [0, 100]$. The speed of the vehicles is set to a fixed value. Different time-window distributions are set for transportation tasks of different sizes, and through a lot of experimental tests, the time-window termination time (T_i) of each node (i) is uniformly and randomly selected from $[300, 1200]$, and then the length of the time window is $L = T_i$, and L is greater than 0. The neural network is stable to the time windows generated in this way. The above setting is only for the case in which there is no special requirement for cargo volume and transportation time.

The parameters of the model are set as shown in Table 1.

Table 1. Model parameter.

Notation	Value	Sense
ζ	128	High-dimensional vector dimension
$\lvert h^t \rvert$	128	Hidden state dimension
β_1	0.01	Reward function weight coefficient
β_2	0.01	Reward function weight coefficients
Batch size	64	Training batch
α	0.001	Learning rate
Dropout	0.04	Neuron inactivation rate

Taking VRPTW-20 as an example, the training set generates 200,000 instances, each iteration batch selects 200 VRPTW-20 as a training batch from the 200,000 instances, and each training batch performs a gradient update to the strategy network and evaluation network. After completing all the instance data as an epoch, each epoch is completed with a validation set to test the model under. The validation set in this study is composed of 2000 instances, and information such as validation-set running time, average total fleet transportation mileage, and average reward are collected to evaluate the model. Setting the iteration at 200 epochs, each iteration completes 1 epoch to validate the validation set.

When 200 iterations are completed, the model training is completed, and we enter the test model stage. The data in the test set and validation set are randomly generated in the same way for 2000 instances to calculate the average fleet mileage, driving time, etc.

In this subsection, DRL algorithms based on a single network, as well as DRL algorithms based on a combined network model, will be compared to simulate and test the same simulated scenario, respectively. In this section, four different scenarios will be set up, and rules will be set to generate four different datasets. In this case, when the scale growth is 0, it means that the size of the rescue points remains unchanged, and when the scale growth is 5, it means that the size of the rescue points grows from 40 to 45 in a randomized insertion manner. The rate of change in the data indicates a 10% oscillation of the parameters of the rescue point, where the parameters include the horizontal coordinate, vertical coordinate, and value of the demand. The oscillation refers to the parameter increasing or decreasing after an action.

4.2. Simulation Results and Model Evaluation

After the model training is completed, the model is tested using a random sampling decoding strategy. Figure 2a–d display the rescue-scheduling plans obtained by solving the model for the 20-point standard dataset, the 40-point scale growth dataset, the 60-point scale growth dataset, and the 100-point scale growth dataset, respectively. The black asterisk represents the initial point, red dots represent rescue points, and the blue line depicts the rescue route of the team.

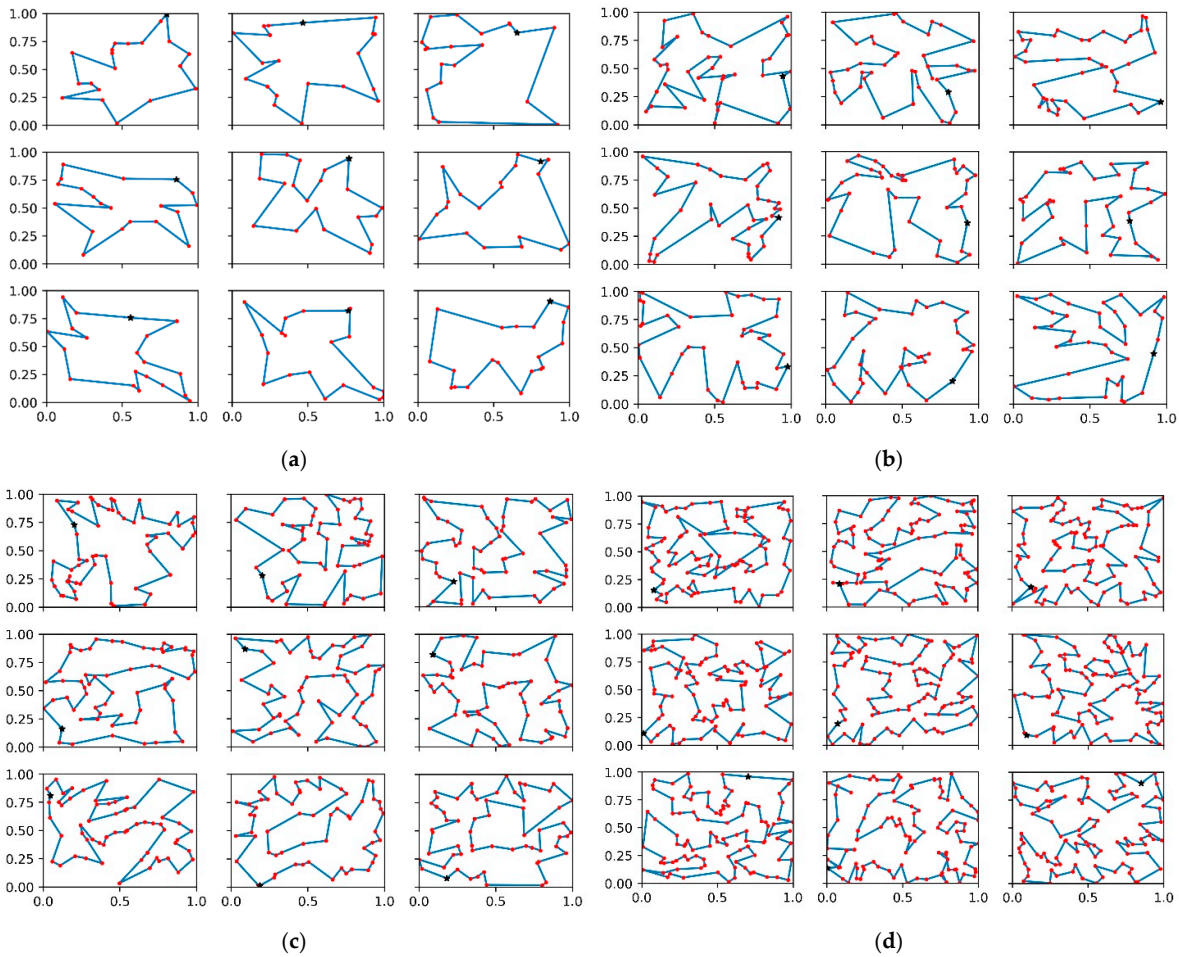

Figure 2. Graphs for (**a**) 20-point standard scheduling program, (**b**) 40-point dynamic growth scale scheduling program, (**c**) 60-point dynamic growth scale scheduling program, and (**d**) 100-point dynamic growth scale scheduling program.

The ACO algorithm, DRL algorithm based on the traditional single network, and DRL algorithm based on combinatorial network model were used to conduct simulation experiments on datasets of different sizes. Each algorithm solved the same dataset 1000 times. The comparison of the average time spent, the average objective function value and the variance is shown in Table 2.

Table 2. Results of the three methods for different sizes and growth datasets.

Size of Rescue Points	Model	Average Time Spent (s)	Average Objective Function Value	Objective Function Variance
20-point standard dataset	Combinatorial network model	17.02	4.36	0
	Traditional single network model	4.24	5.07	0.000299
	ACO	40.4	3.57	0.000324
40-point standard dataset	Combinatorial network model	28.16	5.91	0
	Traditional single network model	7.04	6.57	0.000402
	ACO	272.2	5.30	0.00173
40-point scale growth dataset (10%)	Combinatorial network model	31.04	6.15	0
	Traditional single network model	7.77	6.79	0.000735
	ACO	892.0	5.45	0.00421
60-point scale growth dataset (10%)	Combinatorial network model	53.14	7.43	0
	Traditional single network model	13.29	8.30	0.000686
	ACO	3688.02	6.82	0.00601
100-point scale growth dataset (10%)	Combinatorial network model	96.08	9.50	0.000212
	Traditional single network model	24.01	11.75	0.000679
	ACO	-	-	-

Note: "-" indicates that the algorithm used was not able to obtain results in the allowed time under this scale condition.

The effect of the comparison between the calculation time and the target value is shown in Figures 3 and 4 below.

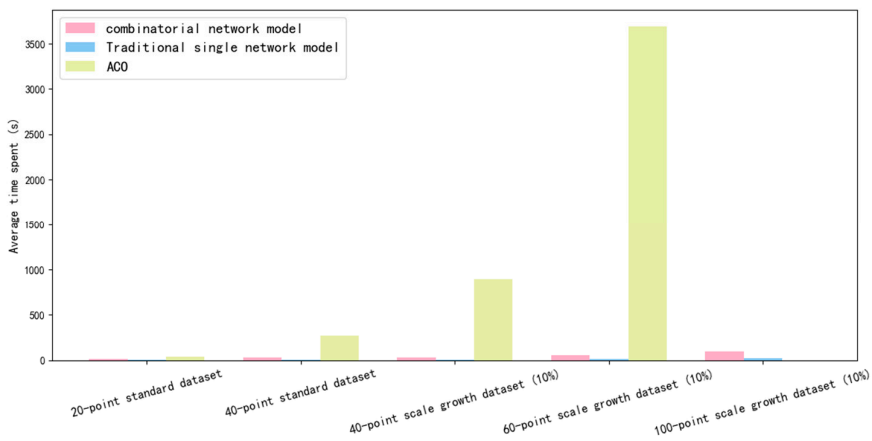

Figure 3. Comparative effect pictures of the calculation time of the three methods at different scales.

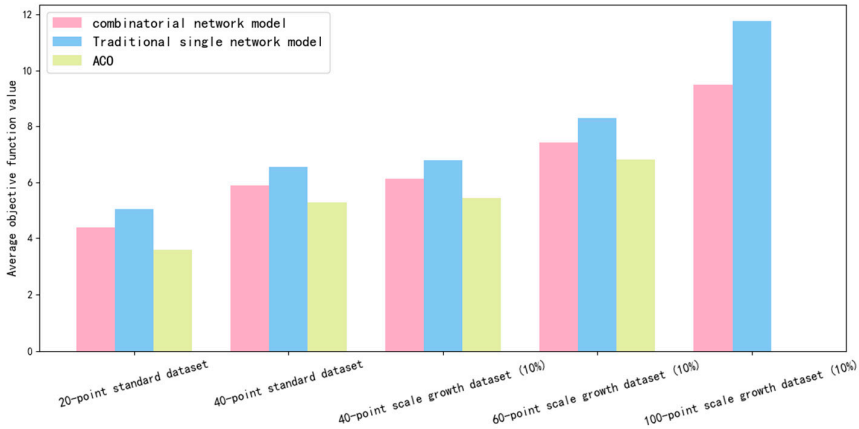

Figure 4. Comparative effect pictures of the target values of the three methods at different scales.

As can be seen from the table, the variance of all three objective functions is at the same level and has a very small value. In the first four experiments, the solution accuracy of the ACO algorithm is higher than that of the DRL algorithm based on the single network model and the combined network model. In terms of computation time, the solution time of the ACO algorithm is higher than that of the remaining two methods as the scale increases; especially in the simulation experiments with 40- and 60-point scale growth datasets, the solution time of the ACO algorithm is 30 and 70 times more than that of the combinatorial network model, which is not in line with the emergency scenario. The solution time of the combined network model is higher than that of the single network model, but it belongs to the acceptable range, which meets the conditions of emergency rescue, and its solution accuracy is slightly higher. Therefore, it can initially be concluded that the DRL algorithm based on the combined network model is better than the DRL algorithm based on the single network model and the ACO algorithm in dealing with the emergency logistics scheduling problem under the conditions of large-scale and dynamic growth.

In order to confirm the conclusions proposed, the authors also set up a 100-point scale growth dataset to conduct simulation experiments. The results show that the ACO algorithm is no longer able to obtain the results within the time limit under this condition, which is not in line with the application scenario of emergency logistics, but the DRL algorithm still has strong stability under this condition. At the same time, the combined network model is higher than the single network model in terms of calculation time but still within the acceptable range, and it can obtain stable results with higher accuracy.

Through the simulation verification of five training sets, the DRL algorithm based on combinatorial network model proposed in this paper can effectively solve EVRP with large-scale and dynamic growing demands, ensuring high accuracy and acceptable solution times. Simulation experiments demonstrate that the proposed method reduces the solution time by 80% compared to the ACO algorithm, although the solution accuracy is reduced by 10%. Compared to the DRL algorithm based on the traditional single-network model, the solution accuracy is improved by almost 20%, although the solution time is slightly increased. Therefore, the authors conclude that the proposed method is superior to the traditional method while maintaining no significant increase in rescue time.

5. Conclusions

In this paper, the authors propose an advanced DRL algorithm based on a combination network model, which uses the actor–critic framework and combines attention networks, pointer networks, and LSTM neural networks to conduct in-depth research on dynamic logistics scheduling problems in emergency rescue scenarios. The authors set up a simula-

tion environment based on standard and dynamic demand. The main research results of this paper are as follows:

(1) This paper proposed a combined neural network model, applying a DRL algorithm that combines pointer networks, LSTM, and attention mechanisms, effectively tackling the dynamic routing problem emergency logistics.

(2) On the basis of (1), it was found through various dynamic simulation datasets that heuristic algorithms perform more stably and accurately in small- and medium-sized training sets. However, in dynamic demand scenarios, the solution time is too long and not suitable for emergency environments. For the two neural network models used, the combined neural network model performs better than a single neural network model when the rescue point scale is large and dynamically growing.

This study examines the dynamic changes in the scale of rescue points in EVRP, validating the superiority of the proposed DRL algorithm in terms of both solution time and accuracy compared to traditional algorithms, as well as its effectiveness in solving real-world problems. However, several limitations should be acknowledged. Firstly, the study assumes that road conditions are favorable during rescue operations. In reality, sudden events may cause road accessibility to be hindered, leading to complex traffic conditions. Therefore, the method proposed in this paper is applicable to scenarios where a sudden event occurs, but the road accessibility between rescue points is not significantly affected, such as the COVID-19 pandemic. Situations like public health emergencies, exemplified by the COVID-19 pandemic, underscore the essential role of emergency logistics in swiftly adapting to evolving conditions. While roads remained navigable, the fluctuating demands for medical supplies, vaccines, and protective equipment necessitated ongoing adjustments in routing. Similarly, scenarios like urban pipeline ruptures, industrial accidents, and controlled evacuations during large-scale events require emergency teams to navigate accessible yet dynamically altering routes. These examples illustrate that although some disasters may impair infrastructure, numerous emergencies still allow for dynamic logistics operations. The model presented herein effectively optimizes real-time routing under such circumstances, reinforcing its practical relevance beyond mere theoretical applications. Secondly, it simplifies the emergency logistics scenario by considering only a single vehicle. This limitation could restrict its application in more complex, real-world situations that often require coordinating multiple vehicles and varied transportation modes.

Future research will tackle these limitations by incorporating new elements, such as pre-positioned standby rescue vehicles to manage the volatile demand. In the simulation validation phase, certain road segments between rescue points could be randomly assumed to be impassable. This approach will allow for a more realistic simulation of dynamic constraints and road blockages. Furthermore, future studies will explore the coordination and optimization between multiple rescue teams, considering factors such as vehicle capacity and real-time traffic data, to enhance the overall efficiency and effectiveness.

Author Contributions: J.Z., conceptualization, writing—review and editing, and visualization; H.X., literature search and writing— original draft; D.L., editing; Q.Y., methodology, writing—original draft, and validation. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Humanities and Social Sciences Fund of the Ministry of Education, grant numbers 20YJCZH225 and 23YJC630109.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Kemball-Cook, D.; Stephenson, R. Lessons in logistics from Somalia. *Disasters* **1984**, *08*, 57–66. [CrossRef] [PubMed]
2. Kundu, T.; Sheu, J.; Kuo, H. Emergency logistics management—Review and propositions for future research. *Transp. Res. Part E Logist. Transp. Rev.* **2022**, *164*, 102789. [CrossRef]
3. Bolaños, C.; Rojas, B.; Salazar-Cabrera, R.; Ramírez-González, G.; de la Cruz, Á.P.; Molina, J.M.M. Fleet management and control system for developing countries implemented with Intelligent Transportation Systems (ITS) services. *Transp. Res. Interdiscip. Perspect.* **2022**, *16*, 100694. [CrossRef]
4. Zhang, J.; Dong, M.; Chen, F.F. A bottleneck Steiner tree based multi-objective location model and intelligent optimization of emergency logistics systems. *Robot. Comput. -Integr. Manuf.* **2013**, *29*, 48–55. [CrossRef]
5. Ju, X.; Su, S.; Xu, C.; Wang, H. Computation offloading and tasks scheduling for the internet of vehicles in edge computing: A deep reinforcement learning-based pointer network approach. *Comput. Netw.* **2023**, *223*, 109572. [CrossRef]
6. Chung, S.H. Applications of smart technologies in logistics and transport: A review. *Transp. Res. Part E Logist. Transp. Rev.* **2021**, *153*, 102455. [CrossRef]
7. Zhang, Q.; Xiong, S. Routing optimization of emergency grain distribution vehicles using the immune ant colony optimization algorithm. *Appl. Soft Comput.* **2018**, *71*, 917–925. [CrossRef]
8. Peng, W.; Wang, D.; Yin, Y.; Cheng, T.C.E. Multi-agent deep reinforcement learning-based truck-drone collaborative routing with dynamic emergency response. *Transp. Res. Part E Logist. Transp. Rev.* **2025**, *195*, 103974. [CrossRef]
9. Braekers, K.; Ramaekers, K.; Van Nieuwenhuyse, I. The vehicle routing problem: State of the art classification and review. *Comput. Ind. Eng.* **2016**, *99*, 300–313. [CrossRef]
10. Sheu, J.B. Dynamic relief-demand management for emergency logistics operations under large-scale disasters. *Transp. Res. Part E Logist. Transp. Rev.* **2010**, *46*, 1–17. [CrossRef]
11. Pillac, V.; Gendreau, M.; Guéret, C.; Medaglia, A.L. A review of dynamic vehicle routing problems. *Eur. J. Oper. Res.* **2013**, *225*, 1–11. [CrossRef]
12. Karp, R.M. Reducibility among combinatorial problems. In *Complexity of Computer Computations*; Miller, R.E., Thatcher, J.W., Eds.; Plenum Press: New York, NY, USA, 1972; pp. 85–103.
13. Lenstra, J.K.; Kan, A.H. Complexity of vehicle routing and scheduling problems. *Networks* **1981**, *11*, 221–227. [CrossRef]
14. Silver, D.; Huang, A.; Maddison, C.J.; Guez, A.; Sifre, L.; Driessche, G.V.D.; Schrittwieser, J.; Antonoglou, I.; Panneershelvam, V.; Lanctot, M. Mastering the game of Go with deep neural networks and tree search. *Nature* **2016**, *529*, 484–489. [CrossRef] [PubMed]
15. Dantzig, G.B.; Ramser, J.H. The truck dispatching problem. *Manag. Sci.* **1959**, *6*, 80–91. [CrossRef]
16. Balinski, M.L.; Quandt, R.E. On an Integer Program for a Delivery Problem. *Oper. Res.* **1964**, *12*, 300–304. [CrossRef]
17. Zhang, X.; Hao, Y.; Zhao, X.; Yuan, X. A novel vehicle path planning method for freight enterprises considering environmental regulation. *J. Clean. Prod.* **2023**, *423*, 138839. [CrossRef]
18. Wang, Z.; Wen, P. Optimization of a Low-Carbon Two-Echelon Heterogeneous-Fleet Vehicle Routing for Cold Chain Logistics under Mixed Time Window. *Sustainability* **2020**, *12*, 1967. [CrossRef]
19. Amiri, A.; Amin, S.H.; Zolfagharinia, H. A bi-objective green vehicle routing problem with a mixed fleet of conventional and electric trucks: Considering charging power and density of stations. *Expert Syst. Appl.* **2023**, *213*, 119228. [CrossRef]
20. Jiang, Y.; Hu, W.; Gu, W.; Yu, Y.; Xu, M. A multi-mode hybrid electric vehicle routing problem with time windows. *Transp. Res. Part E Logist. Transp. Rev.* **2025**, *195*, 103976. [CrossRef]
21. Jie, K.W.; Liu, S.Y.; Sun, X.J. A hybrid algorithm for time-dependent vehicle routing problem with soft time windows and stochastic factors. *Eng. Appl. Artif. Intell.* **2022**, *109*, 104606. [CrossRef]
22. Wang, W.; Li, Y.; Yan, H.; Zhao, W.; Zhao, Q.; Luo, K. A two-phase algorithm for the dynamic time-dependent green vehicle routing problem in decoration waste collection. *Expert Syst. Appl.* **2025**, *262*, 125570. [CrossRef]
23. Wang, Y.; Gou, M.; Luo, S.; Fan, J.; Wang, H. The multi-depot pickup and delivery vehicle routing problem with time windows and dynamic demands. *Eng. Appl. Artif. Intell.* **2025**, *139*, 109700. [CrossRef]
24. Najafi, M.; Eshghi, K.; de Leeuw, S. A dynamic dispatching and routing model to plan/replan logistics activities in response to an earthquake. *Oper. Res. Spektrum* **2014**, *36*, 323–356. [CrossRef]
25. Ceselli, A.; Righini, G.; Tresoldi, E. Vehicle routing problems with different service constraints: A branch-and cut-and-price algorithm. *Networks* **2014**, *64*, 282–291. [CrossRef]
26. Lozano, L.; Duque, D.; Medaglia, A.L. An Exact Algorithm for the Elementary Shortest Path Problem with Resource Constraints. *Transp. Sci.* **2016**, *50*, 348–357. [CrossRef]
27. Elshaer, R.; Awad, H. A taxonomic review of metaheuristic algorithms for solving the vehicle routing problem and its variants. *Comput. Ind. Eng.* **2020**, *140*, 106242. [CrossRef]
28. Li, J.K.; Li, J.Q.; Xu, Y. An improved multiobjective evolutionary algorithm for time-dependent vehicle routing problem with time windows. *Egypt. Inform. J.* **2024**, *28*, 100574. [CrossRef]

29. Salamatbakhsh-Varjovi, A.; Tavakkoli-Moghaddam, R.; Alinaghian, M.; Najafi, E. Robust periodic vehicle routing problem with time windows under uncertainty: An efficient algorithm. *KSCE J. Civ. Eng.* **2018**, *22*, 4626–4634. [CrossRef]
30. Li, Y.; Soleimani, H.; Zohal, M. An improved ant colony optimization algorithm for the multi-depot green vehicle routing problem with multiple objectives. *J. Clean. Prod.* **2019**, *227*, 1161–1172. [CrossRef]
31. Wu, H.; Gao, Y. An ant colony optimization based on local search for the vehicle routing problem with simultaneous pickup–delivery and time window. *Appl. Soft Comput.* **2023**, *139*, 110203. [CrossRef]
32. Teng, Y.; Chen, J.; Zhang, S.; Wang, J.; Zhang, Z. Solving dynamic vehicle routing problem with time windows by ant colony system with bipartite graph matching. *Egypt. Inform. J.* **2024**, *25*, 100421. [CrossRef]
33. Chen, M.; Chen, Y.; Du, Y.; Wei, L.; Chen, Y. Heuristic algorithms based on deep reinforcement learning for quadratic unconstrained binary optimization. *Knowl. -Based Syst.* **2020**, *207*, 106366. [CrossRef]
34. Tang, M.; Zhuang, W.; Li, B.; Liu, H.; Song, Z.; Yin, G. Energy-optimal routing for electric vehicles using deep reinforcement learning with transformer. *Appl. Energy* **2023**, *350*, 121711. [CrossRef]
35. Xin, L.; Song, W.; Cao, Z.; Zhang, J. Step-wise deep learning models for solving routing problems. *IEEE Trans. Ind. Inform.* **2020**, *17*, 4861–4871. [CrossRef]
36. Wang, Y.; Hong, X.; Wang, Y.; Zhao, J.; Sun, G.; Qin, B. Token-based deep reinforcement learning for Heterogeneous VRP with Service Time Constraints. *Knowl. -Based Syst.* **2024**, *300*, 112173. [CrossRef]
37. Zhang, K.; He, F.; Zhang, Z.; Lin, X.; Li, M. Multi-vehicle routing problems with soft time windows: A multi-agent reinforcement learning approach. *Transp. Res. Part C Emerg. Technol.* **2020**, *121*, 102861. [CrossRef]
38. Secomandi, N. Comparing neuro-dynamic programming algorithms for the vehicle routing problem with sto chastic demands. *Comput. Oper. Res.* **2000**, *27*, 1201–1225. [CrossRef]
39. Zhou, C.; Ma, J.; Douge, L.; Chew, E.P.P.; Lee, L.H. Reinforcement learning-based approach for dynamic vehicle routing problem with stochastic demand. *Comput. Ind. Eng.* **2023**, *182*, 109443. [CrossRef]
40. Kamrani, M.; Srinivasan, A.R.; Chakraborty, S.; Khattak, A.J. Applying Markov decision process to understand driving decisions using basic safety messages data. *Transp. Res. Part C Emerg. Technol.* **2020**, *115*, 102642. [CrossRef]
41. Wang, J.; Liu, K.; Li, H. LSTM-based graph attention network for vehicle trajectory prediction. *Comput. Netw.* **2024**, *248*, 110477. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.